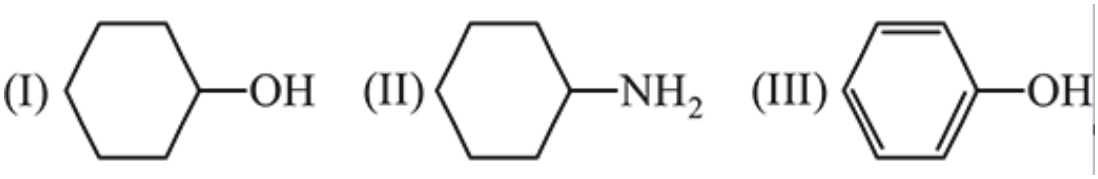
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Arrange in the order of increasing acidity.



(A) $\mathrm{III} < \mathrm{I} < \mathrm{II}$

(B) $\mathrm{I} < \mathrm{III} < \mathrm{II}$

(C) III < II < I

(D) $\mathrm{II} < \mathrm{I} < \mathrm{III}$

Q.2

Heterolysis of carbon-chlorine bond produces

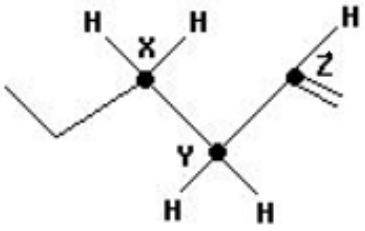
(A) Two free radicals

(B) Two carboniumions

(C) Two carbanions

(D) One cation and one anion

Q.3



Arrange the C-H bods $'x'$, $'y'$ and $'z'$ in decreasing order of their bond dissociation energies homolysis

(A) $\mathrm{Y} > \mathbf{X} > \mathbf{Z}$

(B) $Z > \mathbf{X} > \mathbf{Y}$

(C) $\mathrm{Z} > \mathrm{Y} > \mathbf{X}$

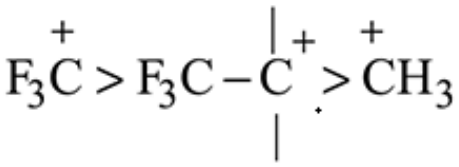
(D) $\mathrm{Y} > \mathbf{Z} > \mathbf{X}$

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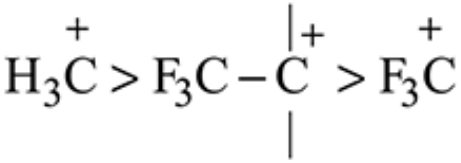
Q.4

Which of the following represents the correct order of stability of the given carbocations?

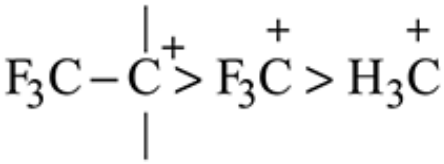
(A)



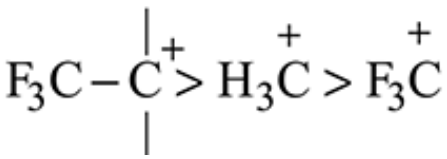
(B)



(C)



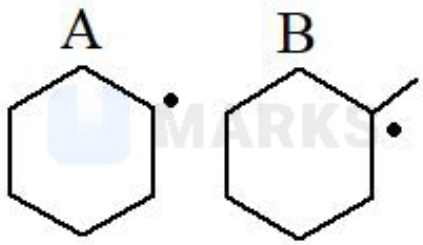
(D)



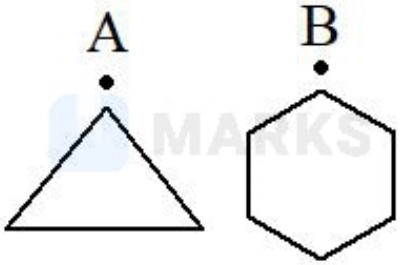
Q.5

In which of the following pairs A is more stable than B?

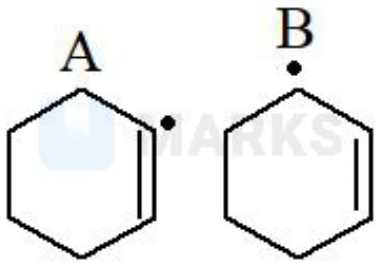
(A)



(B)



(C)

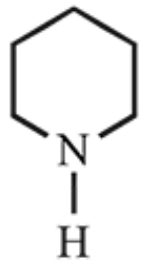


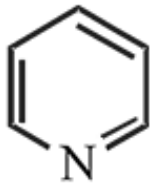
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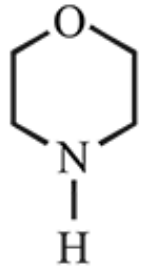
(D) A: $Ph_3C^{\bullet}$, B: $(CH_3)_3C^{\bullet}$

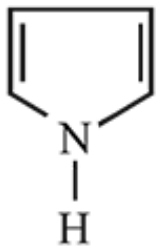
Q.6

In the following compounds

(I) 

(II) 

(III) 

(IV) 

the order of basicity is :

(A) IV > I > III > II

(B) III > I > IV > II

(C) II > I > III > IV

(D) I > III > II > IV

Q.7

Which of the following statements is not correct?

(A) Carbocation posses sextet of electrons.

(B) The order of carbocation stability is :

$$CH_3^+ > (CH_3)_2CH^+ > (CH_3)_3C^+$$

(C) Carbocations have trigonal planar shape

(D) Carbocations are formed by heterolytic cleavage

Q.8

Which is the most stable carbocation?

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(A) iso- propyl cation

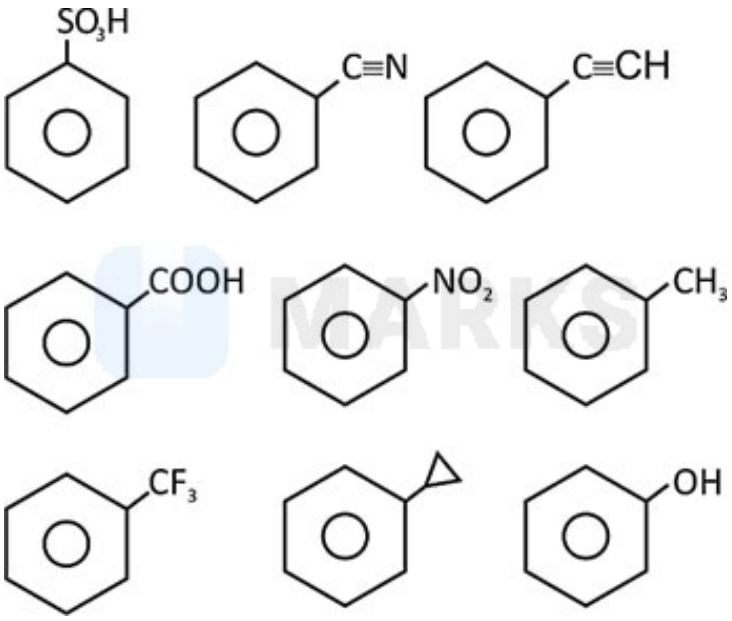
(B) Triphenylmethyl cation

(C) Ethyl cation

(D) n-propyl cation

Q.9

X = number of compounds having -I group directly attached to benzene.



Find the value of X?

(A) 5

(B) 6

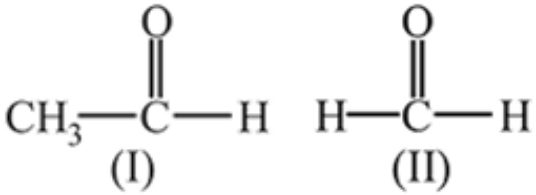
(C) 7

(D) 8

Q.10

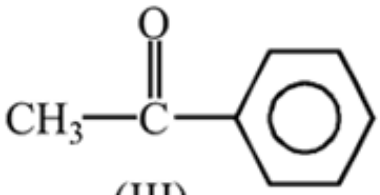
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Correct order of reactivity of following compounds towards Grignard reagent?



(I)

(II)



(III)

(A) $\mathrm{I} > \mathrm{II} > \mathrm{III}$

(B) $\mathrm{II} > \mathrm{I} > \mathrm{III}$

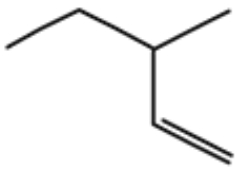
(C) $\mathrm{II} > \mathrm{III} > \mathrm{I}$

(D) $\mathrm{I} > \mathrm{III} > \mathrm{II}$

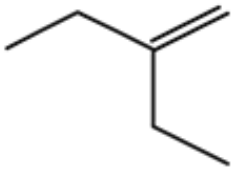
Q.11

Which of the following alkenes is the most stable?

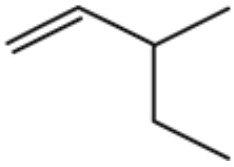
(A)



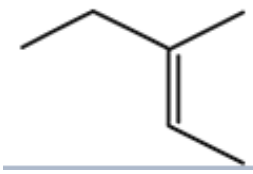
(B)



(C)



(D)



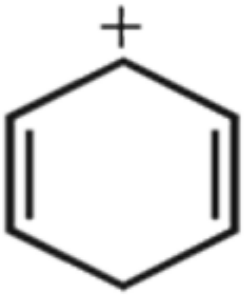
Q.12

Which one amongst the following carbocations is most stable?

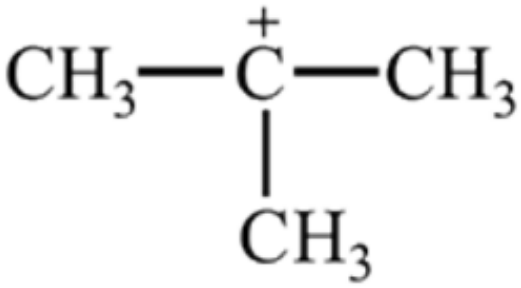
(A) $\mathrm{C_6H_5 - C^+H - C_6H_5}$

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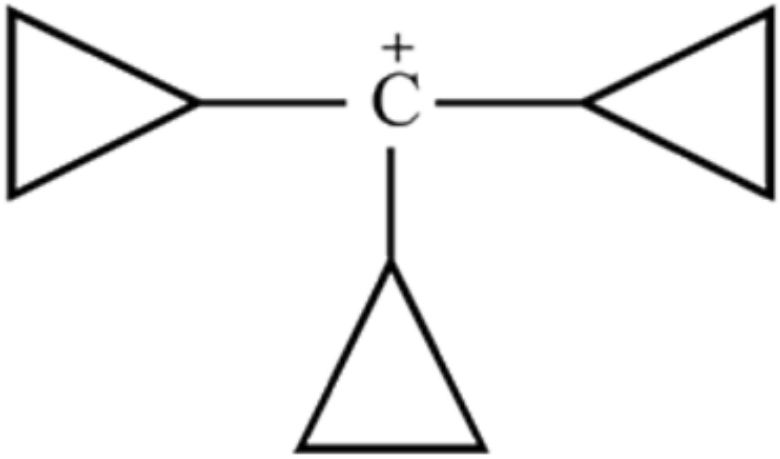
(B)



(C)



(D)



Q.13

In the Friedel-Crafts acylation, the electrophile is

(A) $C_6H_5^+$

(B) $AlCl_3^-$

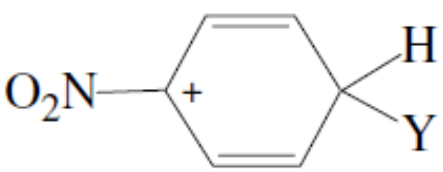
(C) CH_3CO^+

(D) $C_6H_5CH_2^+$

Q.14

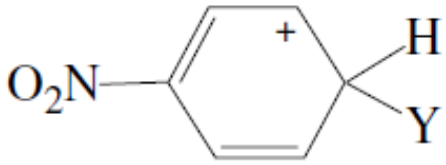
Which of the following carbocations is expected to be most stable?

(A)

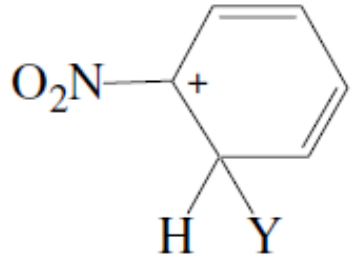


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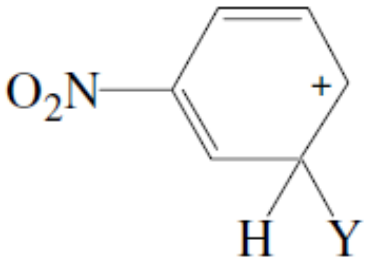
(B)



(C)



(D)



Q.15

Which of the following species is expected to yield maximum percentage of meta substitution product?

(A) $ArCH_3$

(B) $ArCH_2Cl$

(C) $ArCHCl_2$

(D) $ArCCl_3$

Q.16

Arrange the carbanions, $(CH_3)_3\,C^-$, Cl_3C^-, $(CH_3)_2\,CH^-$, $C_6H_5CH_2^-$, in order of their decreasing stability.

(A) $Cl_3C^- > C_6H_5CH_2^- > (CH_3)_2\,CH^- > (CH_3)_3\,C^-$

(B) $(CH_3)_3\,C^- > (CH_3)_2\,CH^- > C_6H_5CH_2^- > Cl_3C^-$

(C) $C_6H_5CH_2^- > Cl_3C^- > (CH_3)_3\,C^- > (CH_3)_2\,CH^-$

(D) $(CH_3)_2\,CH^- > Cl_3C^- > C_6H_5CH_2^- > (CH_3)_3\,C^-$

Q.17

The increasing order of stability of the following free radicals are

(A) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C}$

(B) $(C_6H_5)_3\dot{C} < (C_6H_5)_2\dot{C}H < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$

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(C) $(C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C} < (CH_3)_3\dot{C} < (CH_3)_2\dot{C}H$

(D) $(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_3\dot{C} < (C_6H_5)_2\dot{C}H$

Q.18

A solution of $(-)-1-$ chloro-1-phenylethane in toluene racemises slowly in the presence of a small amount of $SbCl_5$ due to the formation of

(A) Carbanion

(B) Carbene

(C) Carbocation

(D) Free radical

Q.19

Among the following, the strongest base is

(A) $C_6H_5NH_2$

(B) $p-NO_2C_6H_4NH_2$

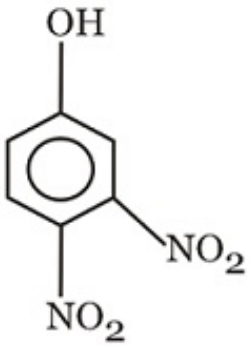
(C) $m-NO_2C_6H_4NH_2$

(D) $C_6H_5CH_2NH_2$

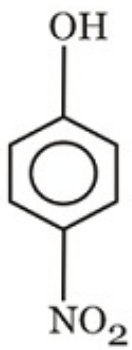
Q.20

Most acidic hydrogen containing compound among the following is

(A)

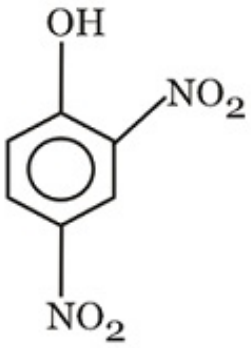


(B)

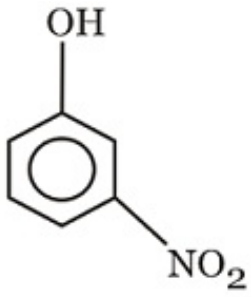


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(C)



(D)



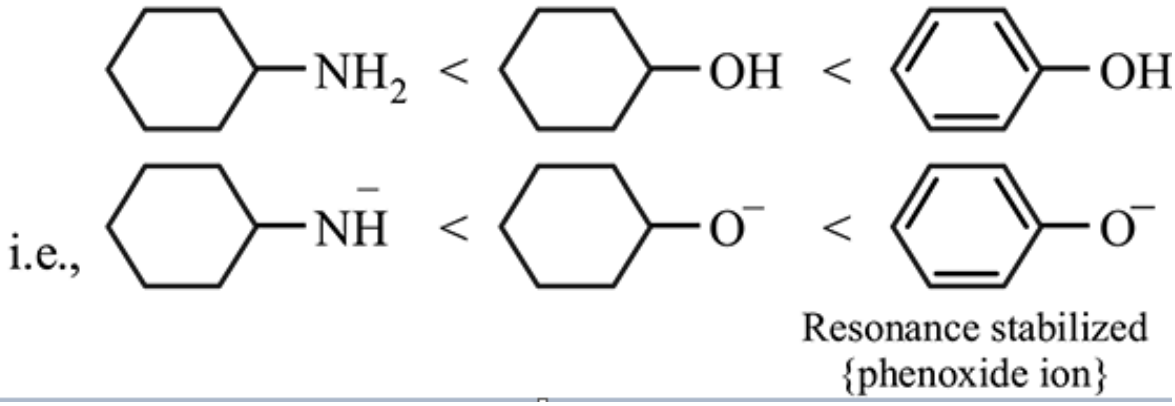
Answers & Solutions

Q.1 Answer:

$\mathrm{II} < \mathrm{I} < \mathrm{III}$

Solution:

Order of Acidic strength $\propto$ stability of conjugate base.



Q.2 Answer:

One cation and one anion

Solution:

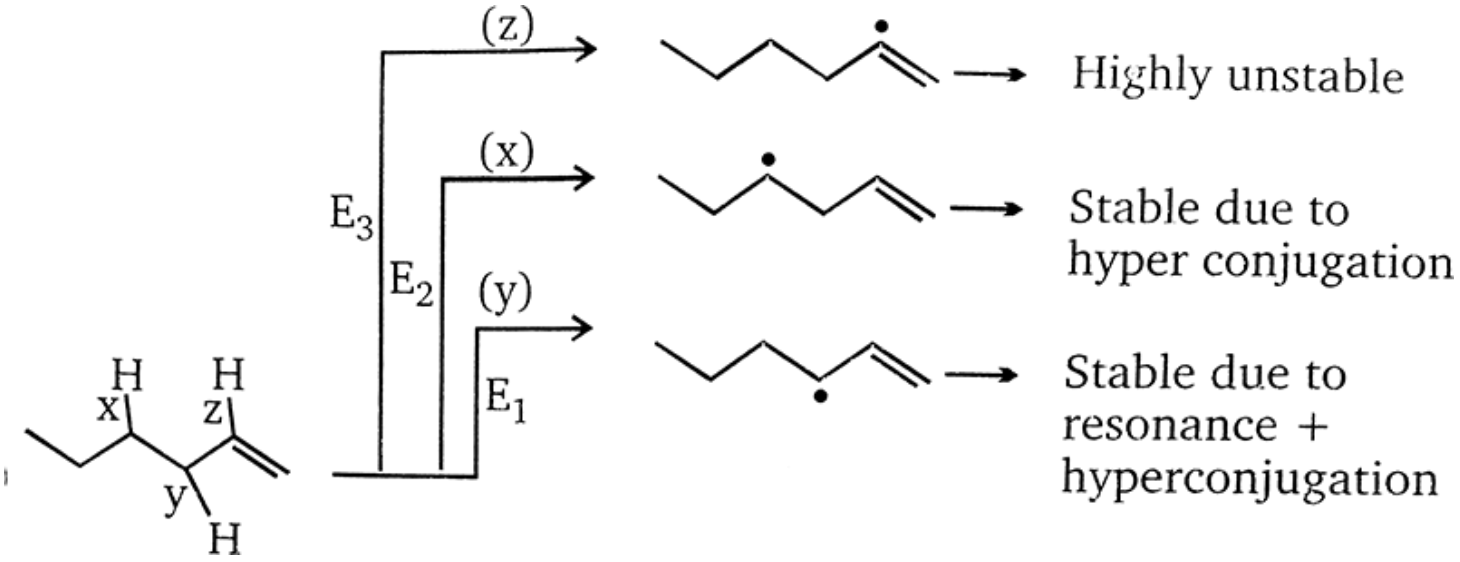
$$\mathrm{C-Cl} \xrightarrow[\text{bond fission}]{\text{Heterolytic}} \mathrm{C^+ + Cl^-}$$

Q.3 Answer:

$Z > \mathbf{X} > \mathbf{Y}$

Solution:

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so bond dissociation energy order is

$E_1 > E_2 > E_3 \quad \text{i. } e, \quad (z) > (x) > (y)$

Q.4 Answer:

$$H_3\overset{+}{C} > F_3C-\overset{+}{\underset{|}{C}} > F_3\overset{+}{C}$$

Solution:

– Igroup destablises carbocation and since inductive effect decreases with increasing length of carbon chain. Therefore (b) is the correct option.

Q.5 Answer:

A B

$Ph_3C^{\bullet}, (CH_3)_3C^{\bullet}$

Solution:

$Ph_3C^{\bullet} > CH_3C^{\bullet}$

In $Ph_3C^{\bullet}e^{-}$ undergo in delocalization while $CH_3C^{\bullet}$ is stable due to the $+I$ effect of $-CH_3$ group, because resonance is a predominant factor so order is above

Q.6 Answer:

$I > III > II > IV$

Solution:

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In I the unshared pair of electrons on N always available for protonation in III due to presence of electronegative O atom the electron density on N is decreased . In II and IV resonance suppresses the basic character.

Q.7 Answer:

The order of carbocation stability is :

$$CH_3^+ > (CH_3)_2CH^+ > (CH_3)_3C^+$$

Solution:

The order of stability of carbocations is :

$$(CH_3)_3\,C^+ > (CH_3)_2\,CH^+ > CH_3^+$$

Q.8 Answer:

Triphenylmethyl cation

Solution:

The correct option is (2): Triphenylmethyl cation.

Stability of carbonium ions follows the order:

tertiary alkyl > secondary alkyl > primary alkyl > methyl

More the number of alkyl groups, the greater the dispersal of positive charge and therefore, more the stability of carbonium ion is observed.

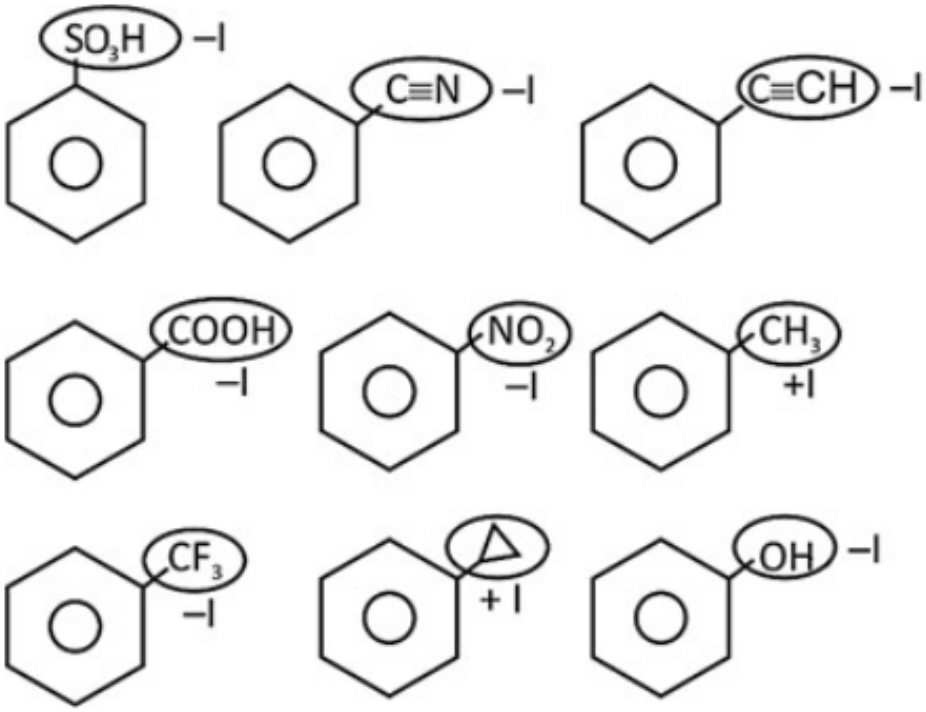
In triphenylmethyl cation, dispersal of positive charge increases with the increase in the number of benzene ring. Hence, it is the most stable carbonium ion.

Q.9 Answer:

7

Solution:

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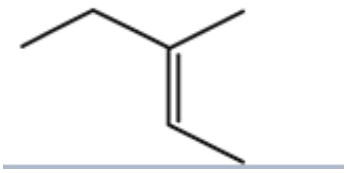
Q.10 Answer:

$\mathrm{II} > \mathrm{I} > \mathrm{III}$

Solution:

CH_3 **group give** $+M$ **and** $+I$ **but** $+M$ **work only at ortho and para position. Reactivity** $\propto$ **positive charge on carbonyl carbon so that's why reactivity** $\mathrm{II} > \mathrm{I} > \mathrm{III}$**.**

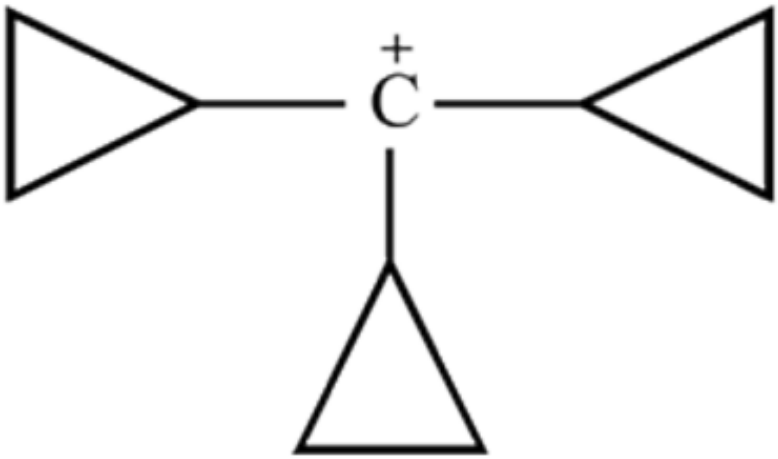
Q.11 Answer:



Solution:

Stability of Alkene $\propto$ **No. of H-atoms or more hyperconjugation.**

Q.12 Answer:



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Solution:

very stable carbocation.

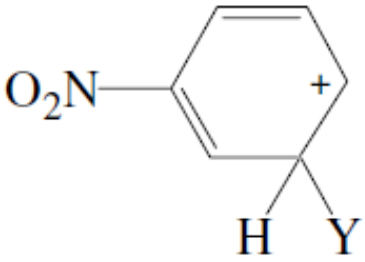
Q.13 Answer:

CH_3CO^+

Solution:

The electrophile is CH_3CO^+.

Q.14 Answer:



Solution:

NO_2 group is benzene deactivator, it is meta directing.

Q.15 Answer:

$ArCCl_3$

Solution:

CCl_3 has a maximum electron attracting power.

Q.16 Answer:

$Cl_3C^- > C_6H_5CH_2^- > (CH_3)_2\,CH^- > (CH_3)_3\,C^-$

Solution:

The carbanion $(CH_3)_3\,C^-$ is expected to be least stable due to electron-releasing nature of methyl groups. The carbanion Cl_3C^- is expected to be most stable due to electron-attracting nature of chlorine atoms Keeping only these two facts in mind, the choice (a) seems to be correct.

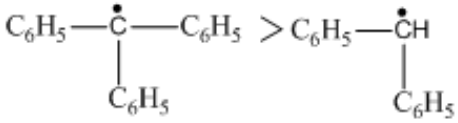
Q.17 Answer:

$(CH_3)_2\dot{C}H < (CH_3)_3\dot{C} < (C_6H_5)_2\dot{C}H < (C_6H_5)_3\dot{C}$

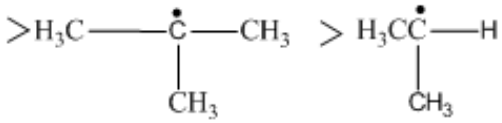
Solution:

Order of free radicals stability:

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Highly stable by delocalisation



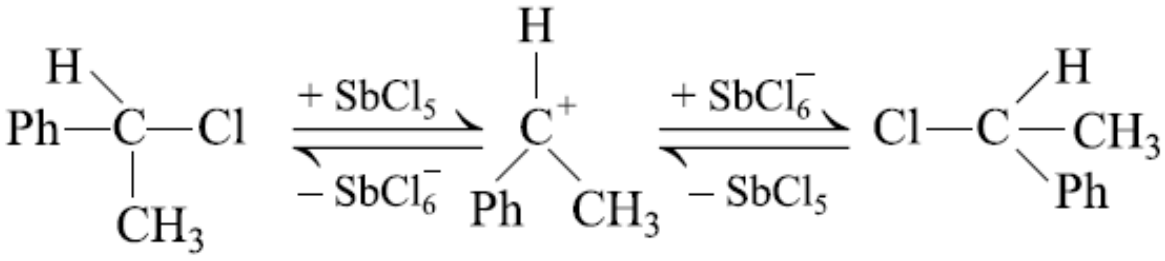
9-hyperconjugative hydrogens and +*I* effect

Q.18 Answer:

Carbocation

Solution:

The reaction proceeds as follows.



There is a formation of intermediate carbocation.

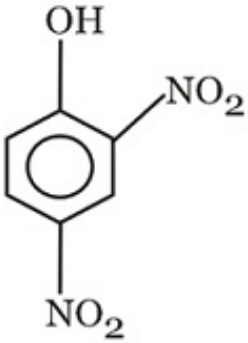
Q.19 Answer:

$C_6H_5CH_2NH_2$

Solution:

$C_6H_5CH_2NH_2$ **is an aliphatic amine. It is most basic amongst the given compounds.**

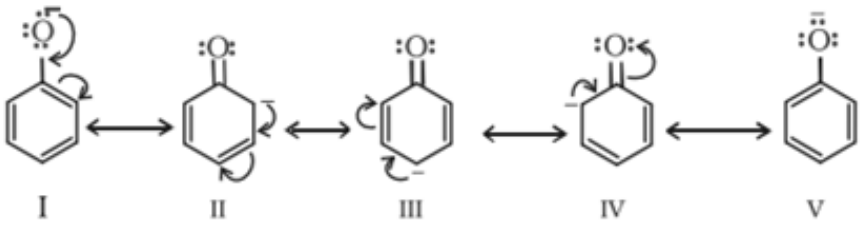
Q.20 Answer:



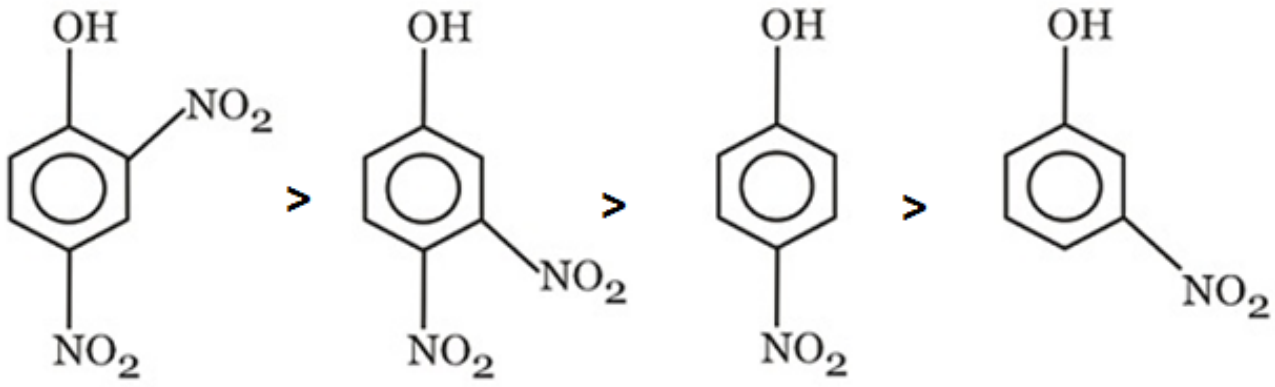
Solution:

In substituted phenols, the presence of electron-withdrawing groups such as nitro group enhances the acidic strength of phenol. This effect is more pronounced when such a group is present at ortho and para positions. It is due to the effective delocalisation of negative charge in phenoxide ion.

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Also, the acidity will increase with the number of electron-withdrawing groups. Thus, the order of acidity of the given compounds is



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